

Class #2 - Characteristic Equation



1. Def: The equation $\det(A - \lambda I) = 0$ is called the **characteristic equation** and the polynomial $\det(A - \lambda I)$ is called the **characteristic polynomial**.

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a \cdot e \cdot i + b \cdot f \cdot g + c \cdot d \cdot h - g \cdot e \cdot c - a \cdot f \cdot h - b \cdot d \cdot i$$

2. Example: $A = \begin{bmatrix} 3 & -5 & 1 \\ 1 & 0 & 2 \\ 0 & -1 & 1 \end{bmatrix}$. Find all eigenvalues of A .

Note that λ is an eigenvalue of $A \Leftrightarrow \det(A - \lambda I) = 0$, i.e., λ is an eigenvalue of $A \Leftrightarrow \lambda$ is a root of the **Characteristic polynomial**.

Have to find λ such that $\det(A - \lambda I) = 0$.

$$\lambda \approx 1.6294, 1.853 \pm 2.1754i$$

$$\Leftrightarrow \det \begin{bmatrix} 3-\lambda & -5 & 1 \\ 1 & -\lambda & 2 \\ 0 & -1 & 1-\lambda \end{bmatrix} = 0$$

$$\Leftrightarrow (-1)^{1+1} (3-\lambda) \begin{vmatrix} -\lambda & 2 \\ -1 & 1-\lambda \end{vmatrix} + (-1)^{2+1} (1) \begin{vmatrix} -5 & 1 \\ -1 & 1-\lambda \end{vmatrix} = 0$$

$$\Leftrightarrow -\lambda^3 + 4\lambda^2 - 10\lambda + 10 = 0 \Leftrightarrow \lambda^3 - 4\lambda^2 + 10\lambda - 10 = 0$$

3. Def: The number of times that certain λ occurs as a root of the characteristic polynomial is called the algebraic multiplicity of λ .

4. Example: Find all eigenvalues of an $n \times n$ identity matrix and state their algebraic multiplicities.

$$\det(I - \lambda I) = \det \begin{bmatrix} 1-\lambda & & 0 \\ & 1-\lambda & \\ 0 & & \ddots & 1-\lambda \end{bmatrix}$$

$$= (1-\lambda)^n = 0$$

$\lambda = 1$ is the only eigenvalue of I

$n \times n$ with algebraic multiplicity n .
 $A - \lambda I$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Fundamental Theorem of Algebra (pf in Complex Analysis): A degree n polynomial has exactly n roots in the complex plane.

5. Note that for a real $n \times n$ matrix A ,

(a) the characteristic polynomial has the degree n . So, A can have exactly n eigenvalues (real or complex), and A can have at most n real eigenvalues.

① Attendance

② Participation quiz

③ Class #2 #3 ~

④ Quiz #1 Mon 9/14 (3:00 - 3:10 pm)
pp #1 ~ #3

$$A \in \mathbb{R}^{4 \times 4}$$

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & i & 0 \\ 0 & 0 & 0 & -i \end{bmatrix}$$

also, multiplicity of $\lambda=1$ is 2.

① $\det(A - \lambda I) = (\lambda - 1)^2 (\lambda - i)(\lambda + i)$
Name: Char. Poly.

② $\det(A - \lambda I) = 0$ Name: Char. Eqn

③ All eigenvalues of A : 1, i, -i

④ $\text{Nul}(A - 1 \cdot I)$ Name: Eigenspace corresponding to $\lambda=1$

⑤ $\dim \text{Nul}(A - 1 \cdot I)$ Name: geo. multiplicity of $\lambda=1$
 $= \text{nullity}(A - 1 \cdot I)$

⑥ Because $\det(A - 1 \cdot I) = \boxed{0}$, $A - 1 \cdot I$ is not invertible.
(nonsingular: invertible, singular: not invertible.)

⑦ Because of ⑥, $\text{nullity}(A - 1 \cdot I) \geq \boxed{1}$.

⑧ $\text{Nul}(A - 1 \cdot I) = \left\{ x \in \mathbb{R}^4 \mid x = \begin{bmatrix} t \\ -t \\ 0 \\ t \end{bmatrix}, t \in \mathbb{R} \right\}$

An Eigenvector of A corresponding to eigenvalue 1 :

$$x = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \end{bmatrix}$$

⑨ Geometric multiplicity of $\lambda=1$ is 1.
In ⑧

(b) (A is invertible) $\Leftrightarrow$ every eigenvalue of A is nonzero.

$\det(A) = \det(A - 0 \cdot I) \neq 0$ | $\lambda = 0$ is eigenvalue of $A \Leftrightarrow \det(A - 0 \cdot I) = 0$

6. Suppose A is row equivalent to B . Do they have the same set of eigenvalues?

Example: $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$

Since A not inv.,
 $\lambda = 0$ is an eigenvalue.

A not invertible (singular)
 $\Leftrightarrow \lambda = 0$ is an eigenvalue of A

$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$
 $r_2 \leftarrow r_2 - 2r_1$

B is triangular.
So eigenvalues of B are 0 and 1.

Qn: A also has 0, 1 as eigenvalues??

Check: $\det(A - 1 \cdot I)$

$= \det \begin{bmatrix} 0 & 2 \\ 2 & 3 \end{bmatrix} = -4 \neq 0$

$\therefore \lambda = 1$ is not an eigenvalue of A .

Fill in the blanks:

Two Row equivalent matrices don't have the same set of eigenvalues.

(a) A and its row reduced matrix do not have the same set of eigenvalues.

(b) But, 0 is an eigenvalue of $A \Leftrightarrow$ 0 is an eigenvalue of its row reduced matrix.

There is a unpivot column.
Its reduced matrix will also have a unpivot column.

$\vec{x}_3 = 2 \cdot \vec{x}_1 + 0 \cdot \vec{x}_2$
 $(\lambda - 1)^2 (\lambda - 3) (\lambda - 4) = 0$
 $n = 6$
 $\lambda_1 = 1 \quad \vec{x}_1$
 $\lambda_2 = 3 \quad \vec{x}_2$
 $\lambda_3 = 4 \quad \vec{x}_3$
 $k = 3$

7. Let A be a real $n \times n$ matrix, and let $\lambda_1, \lambda_2, \dots, \lambda_k$ where $k \leq n$ are real distinct eigenvalues of A . Let $\vec{x}_i$ be an eigenvector corresponding to λ_i for $i = 1, \dots, k$. Then, the set $\{\vec{x}_1, \dots, \vec{x}_k\}$ are L.I. set in $\mathbb{R}^n$. In other words, eigenvectors corresponding to distinct eigenvalues are L.I.

Proof: Suppose $\{\vec{x}_1, \dots, \vec{x}_k\}$ is L.D.

Note: $\vec{x}_i \neq 0$ for all i (eigenvectors)

There exists $\vec{x}_j$ s.t.

$\vec{x}_j = c_1 \vec{x}_1 + \dots + c_{j-1} \vec{x}_{j-1}$
with $\{\vec{x}_1, \dots, \vec{x}_{j-1}\}$ L.I.

$\vec{x}_5 = 0 \cdot \vec{x}_1 + 0 \cdot \vec{x}_3 + 2 \cdot \vec{x}_2 + 0 \cdot \vec{x}_4$

In (*),

① $A \vec{x}_j = A(c_1 \vec{x}_1 + \dots + c_{j-1} \vec{x}_{j-1}) = c_1(A \vec{x}_1) + \dots + c_{j-1}(A \vec{x}_{j-1})$
 $= c_1(\lambda_1 \vec{x}_1) + \dots + c_{j-1}(\lambda_{j-1} \vec{x}_{j-1})$

② $\lambda_j \vec{x}_j = \lambda_j (c_1 \vec{x}_1 + \dots + c_{j-1} \vec{x}_{j-1})$
 $= c_1 \lambda_j \vec{x}_1 + \dots + c_{j-1} \lambda_j \vec{x}_{j-1}$
 $= c_1 (\lambda_j \vec{x}_1) + \dots + c_{j-1} (\lambda_j \vec{x}_{j-1})$

Since ① = ②: $c_1 (\lambda_1 - \lambda_j) \vec{x}_1 + \dots + c_{j-1} (\lambda_{j-1} - \lambda_j) \vec{x}_{j-1} = 0$

Because $\{\vec{x}_1, \dots, \vec{x}_{j-1}\}$ is L.I.,

$c_i (\lambda_i - \lambda_j) = 0$ for all $i=1, 2, \dots, j-1$.

Because $\lambda_1, \dots, \lambda_k$ are all distinct,
 $\lambda_i - \lambda_j \neq 0 \quad \forall i=1, 2, \dots, j-1$.

$\Rightarrow c_i = 0$
 $i=1, 2, \dots, j-1$

$\Rightarrow \vec{x}_j = 0$, but $\vec{x}_j$ is an eigenvector!
(*) Contradiction.

2x2

Suppose $A = \begin{bmatrix} 0 & 3 \\ 0 & 1 \end{bmatrix}$.

eigenvalues of A are 0, 1.

Char. poly. $(\lambda - 0)(\lambda - 1)$

Char. Eqn. $(\lambda)(\lambda - 1) = 0$

alg. mult. of $\lambda = 0$ is 1.
of $\lambda = 1$ is 1.

will show $\text{Nul}(A - 0 \cdot I)$

and $\text{Nul}(A - 1 \cdot I)$ are L.I.

will show: $\text{Nul}(A - 1 \cdot I)$ and $\text{Nul}(A - 0 \cdot I)$ are L.I.

$(\lambda - 1)^2 (\lambda - 0) = 0$

$\dim \text{Nul}(A - 1 \cdot I) = (\text{geo. mult. of } \lambda = 1)$

is 2. There are 3 L.I. eigenvectors corresponding to $\lambda = 1$.

$\dim \text{Nul}(A - 0 \cdot I) \geq 1$
($\det(A - 0 \cdot I) = 0$) $\Rightarrow$ There is a non-trivial soln to $(A - 0 \cdot I)x = 0$.

We get total of 4 L.I. vectors in $\mathbb{R}^3$.

Contradiction!

$\therefore \dim \text{Nul}(A - \lambda I) < \text{alg. mult. of } \lambda$
 $1 \leq = \text{geo. mult. of } \lambda$